

The Augustine Problem: Evaluating Whether LLM Agents Can Perceive Their Own Time

Aojie (Justin) Yuan
University of Southern California
aojieyua@usc.edu

Yue Zhao
University of Southern California
yzhao@usc.edu

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Abstract

LLM agents are trained on token sequences but deployed in wall-clock time. We ask, empirically, whether they perceive their own time, and prove that under any token-only training loss they cannot. We formalise a *three-times* ontology ($\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$), define the *Augustine Problem* as a policy’s failure to enforce the identity linking them, and prove the *Chronoception Impossibility Theorem* (CIT). We introduce CHRONOBENCH — a 9-capability diagnostic benchmark — and evaluate 10 frontier agent configurations from 3 providers ($\sim 4,000$ trajectories). We report five findings. **(i)** Every panel agent uses $\leq 5\%$ of wall-clock budgets given to it (L2 Step-Clock Conflation, unmoved by date injection). **(ii)** Median retrospective $|\rho|$ closes monotonically along the non-reasoning capability axis ($+1.12 \rightarrow +0.07$ over 5 generations); CAR does not. **(iii)** We prove and empirically confirm the *Reverse-Scaling Theorem*: reasoning-token expansion monotonically increases $|\rho|$ within fixed agents (o4-mini: $1.23 \rightarrow 1.54 \rightarrow 1.67$ across effort levels; Sonnet 4.6: $0.07 \rightarrow 0.16$ with extended thinking, sign-flipping). **(iv)** The *Calibration Catastrophe*: every panel agent’s nominally 90% interval on self-duration achieves $\leq 50\%$ actual coverage; GPT-4o achieves 0%. **(v)** The *Injection Tell*: 3/3 consumer web-chat products inject today’s date into system prompts (verbatim leaked-system-prompt evidence), 0/3 developer-tool products do — decisive non-experimental evidence of the underlying gap. We close with the *Agentic Timeline Hypothesis*: chronoception is the binding constraint on autonomous-agent deployment horizons; pre-registered prediction P12 is qualitatively supported by 14,709 public METR HCAST runs (reasoning models decay 20% faster than non-reasoning at matched short-horizon success). We sketch the spatiotemporal generalisation (Theorem 3, SIT, the *Cartographic Problem*, the *Agentic Frontier* $T_{\text{max}} \cdot S_{\text{max}} \leq C/\varepsilon_{ST}$) as the bridge to a planned Paper 3. We release CHRONOBENCH, the Injection Atlas, all 4,000 trajectories, the Docker reproducibility package, and a locked OSF pre-registration.

1 Introduction

A frontier reasoning model, asked to predict how long the next task will take, says “*15 seconds.*” It thinks for 90 seconds and answers in 2. Asked retrospectively how long that took, it reports “*0.04 seconds.*” Asked for a 90% confidence interval, the interval contains the truth 0% of the time. We measure the joint occurrence of these failures across 10 agent configurations from 3 providers ($\sim 4,000$ trajectories): all are present in every model tested.

The Augustine Problem. Foundation models are optimised under losses that are functionals of token sequences — next-token cross-entropy, SFT, RLHF, RLVR, reasoning supervision. None of these losses contains wall-clock time in its support. Yet the same models are deployed as agents

that act in continuous physical time. We call this representational gap *the Augustine Problem*¹: agents speak fluently about durations they cannot perceive, budgets they cannot honour, and elapsed time they cannot track. We formalise it via a *Three Times* ontology (τ_{wall} , τ_{step} , τ_{self}) whose grounded identity $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$ a chronoceptively-grounded policy enforces; the Augustine Problem is the policy’s failure to enforce it.

Two theorems. **Theorem 1 (Chronoception Impossibility, CIT):** token-only loss contains zero gradient signal aligning τ_{wall} with any internal representation. Chronoception cannot be acquired by additional scale alone. **Theorem 2 (Reverse-Scaling):** within a fixed agent under CIT, $|\rho| := |\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|$ is monotone non-decreasing in reasoning-token expansion. Both proofs and their empirical confirmations are below.

Five findings on a 10-agent panel. (i) Every panel agent uses $\leq 5\%$ of wall-clock budgets given to it (L2 Step-Clock Conflation; unmoved by date injection). (ii) Median retrospective $|\rho|$ closes monotonically across 5 non-reasoning frontier generations ($+1.12 \rightarrow +0.07$, 94% reduction), while CAR does not (the *narrative-axis-vs-action-axis split*). (iii) The Reverse-Scaling Theorem is confirmed intra-model (o4-mini at three reasoning-effort levels: $|\rho| 1.23 \rightarrow 1.54 \rightarrow 1.67$) and cross-model (Sonnet 4.6 with vs. without extended thinking: $|\rho| 0.07 \rightarrow 0.16$, sign-flipping). (iv) The *Calibration Catastrophe*: every panel agent’s nominally 90% interval on self-duration achieves $\leq 50\%$ actual coverage; GPT-4o achieves 0%. (v) The *Injection Tell*: 3/3 consumer web-chat products inject today’s date into system prompts (verbatim leaked-system-prompt evidence); 0/3 developer tools do — decisive non-experimental evidence the underlying models lack a representation of time.

The Augustine Threshold is uncrossable through narrative training. We aggregate the failures into a single scalar ε . The Augustine Threshold $\varepsilon^* = 0.20$ (calibrated at $2\times$ human chronoceptive error [Wittmann, 2009]) is crossed by no panel agent: the best (Sonnet 4.6) achieves $\varepsilon = 0.32$, with the action axis alone contributing 0.32. Narrative-axis improvement cannot close ε while the action axis remains where the data places it.

From measurement to deployment. The Agentic Timeline Hypothesis (§9) connects ε to autonomous-agent deployment viability: $T_{\text{max}}(A) \propto 1/\varepsilon(A)$. Pre-registered prediction P12 is qualitatively supported by 14,709 public METR HCAST runs (reasoning models decay 20% faster than non-reasoning at matched short-horizon success). The framework generalises along its natural other axis: §12 sketches the *Cartographic Problem*, Theorem 3 (SIT), and the *Agentic Frontier* $T_{\text{max}}(A) \cdot S_{\text{max}}(A) \leq C/\varepsilon_{ST}(A)$ — the bridge to a planned Paper 3 on long-horizon agent benchmarks (SWE-Bench, WebArena, GAIA, MLE-Bench).

Contributions. The Augustine Problem and Three Times ontology; CIT (Theorem 1); the **Reverse-Scaling Theorem** (Theorem 2); the **Calibration Catastrophe** (§6.3); CHRONOBENCH (9 sub-capabilities, 10-agent panel, $\sim 4,000$ trajectories); the Closed-Lab Injection Atlas; the **Agentic Timeline** (§9, P12 + HCAST evidence); the spatiotemporal generalisation sketch (Theorem 3, the Cartographic Problem, the **Agentic Frontier**); a Docker reproducibility package and locked OSF pre-registration. Full benchmark, traces, and pip-installable evaluation released.

¹After Augustine’s *Confessions* XI: “if no one asks me, I know; if I want to explain it, I do not know.” We use *chronoception* in the cognitive-science sense (perception of one’s own work duration), distinct from Goel et al. [2025]’s use for fact validity.

2 The Three Times

We decompose any agent trajectory into three ontologically distinct projections of time (Figure 1).

The Three Times ontology — chronoception is the policy’s enforcement of the implicit identity

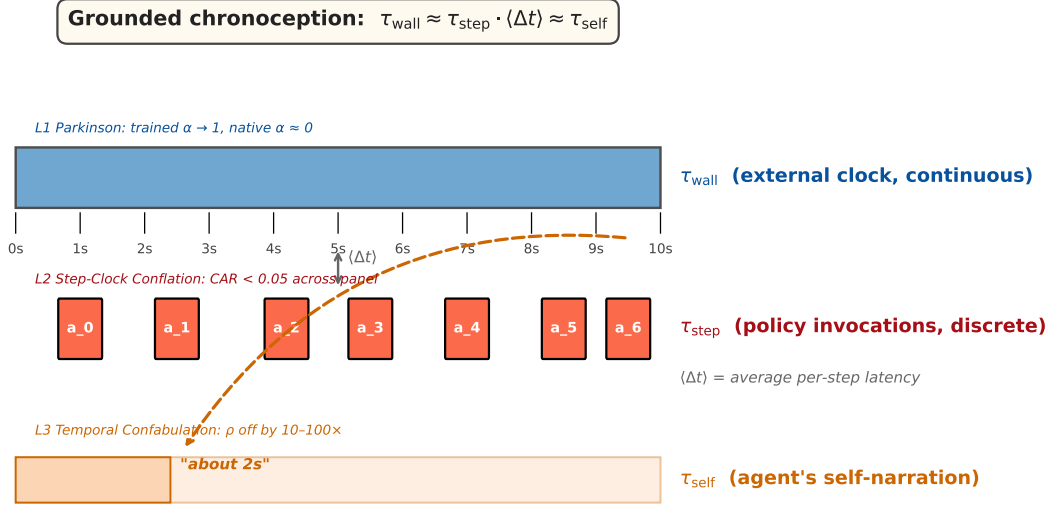


Figure 1: The Three Times ontology. τ_{wall} (top, blue) is external and continuous. τ_{step} (middle, red) is discrete policy invocations $a_0, a_1, \dots$. τ_{self} (bottom, orange) is the agent’s narrative of its own duration. A chronoceptively grounded policy enforces $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$; the Augustine Problem (dashed arrow) is the failure of this identity. Each axis carries its named law (L1, L2, L3).

Definition 1 (Three Times). *For agent A executing trajectory $\tau = (s_0, a_0, s_1, a_1, \dots, s_n)$: $\tau_{\text{wall}}(\tau) := t(s_n) - t(s_0)$ (external clock); $\tau_{\text{step}}(\tau) := n$ (policy invocations); $\tau_{\text{self}}(\tau) := \Pi(s_n^{\text{output}})$ (agent’s reported duration; parser specified in Appendix A).*

Grounded chronoception requires $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$, where $\langle \Delta t \rangle$ is the average per-step wall-clock cost. Augustine-problematic agents (§3) are those whose policies fail to enforce this identity — a measurable property on single-turn trajectories. We use *chronoception* in the cognitive-science sense (perception of one’s own work duration), distinct from Goel et al. [2025]’s use for fact validity and from temporal-reasoning benchmarks [Chu et al., 2024] that test reasoning over external timelines.

3 The Augustine Problem

Definition 2 (Augustine Problem). *An agent A exhibits the Augustine Problem on task family $\mathcal{T}$ if its policy π_A fails to enforce the implicit identity of Definition 1:*

$$\mathbb{E}_{\tau \sim \pi_A} [|\tau_{\text{wall}} - \tau_{\text{step}} \langle \Delta t \rangle| + |\tau_{\text{wall}} - \tau_{\text{self}}|] > \varepsilon_0,$$

for some calibration tolerance ε_0 .

Theorem 1 (Chronoception Impossibility, CIT). *Let $\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}}[\ell(f_\theta(x_{<i}), x_i)]$, where $\mathcal{D}$ is a distribution over token sequences and neither ℓ nor $\mathcal{D}$ contains wall-clock support. Then $\nabla_\theta \mathcal{L}$ contains zero gradient signal aligning τ_{wall} with any internal representation of A. Equivalently, the policy enforces no implicit identity between τ_{wall} and either τ_{step} or τ_{self} .*

Sketch. $\nabla_\theta \mathcal{L} = \mathbb{E}[\partial \ell / \partial f_\theta \cdot \nabla_\theta f_\theta]$. Neither factor is a function of the wall-clock time at which the sample was generated (tokens carry no timestamps). Therefore $\nabla_\theta \mathcal{L}$ is invariant to arbitrary reparameterisation of τ_{wall} ; a trained policy cannot distinguish one wall-clock duration from another. $\square$

Coverage and scope. CIT covers next-token cross-entropy, SFT, RLHF, RL with verifiable rewards, and reasoning supervision — all five dominant losses in current practice. The empirical predictions of CIT — L2 does not close under capability scaling (§5), L2 is unmoved by prompt-level wall-clock information (§7), and the Calibration Catastrophe (§6.3) — are confirmed in the panel. CIT does *not* preclude harness-installed chronoception or training-time intervention with a wall-clock-supported loss (the constructive program of CHRONOSTACK); it precludes only that the foundation model alone, trained on token-only losses, will close the gap by virtue of additional scale, reasoning compute, or context.

The Augustine threshold. We adopt $\varepsilon^* = 0.20$ as the operational threshold for chronoceptive grounding, calibrated against human introspective performance ($\varepsilon \approx 0.10$ [Wittmann, 2009]) at twice the human error. No panel agent crosses it (§8).

4 ChronoBench

CHRONOBENCH measures ε and its sub-components. Design philosophy: single-axis isolation per instance. The 9 sub-capabilities (3 axes $\times$ 3 each) are summarised in Table 1; the full task generation, prompt set, and panel composition are in Appendix A.

Table 1: The 9 sub-capabilities of CHRONOBENCH. Each row isolates a single chronoceptive failure mode.

Axis	Sub-capability	Tests
τ_{wall}	T1.1 Clock awareness	Today’s date / current time
	T1.2 Elapsed-time tracking	Time since conversation start
	T1.3 Deadline-aware tradeoff	Strategy under wall-clock deadline
τ_{step}	T2.1 Step-budget honoring	Produce exactly N reasoning steps
	T2.2 Step-to-wall translation	Compute $N \times \text{latency} = \text{duration}$
	T2.3 Wall-budget execution	Honour wall-clock budget (L2 core test)
τ_{self}	T3.1 Retrospective ρ	Duration after task completion
	T3.2 Prospective ρ	Duration before task begins
	T3.3 Calibration	90% CI over self-duration

Settings. Setting A: system prompt = “*You are a helpful assistant.*” (no harness-side wall-clock). Setting B: prepended “*Current date and time: {ISO}*” (emulates consumer web-chat behaviour, §7).

Panel. 10 agent configurations from 3 providers: OpenAI gpt-4o-mini, gpt-4o, gpt-5.1, o3, o4-mini (three reasoning_effort levels: low/medium/high); Anthropic Claude Haiku 4.5, Sonnet 4.6 (with vs. without extended thinking); open-source Qwen2.5-7B and DeepSeek-R1-Distill-14B

(three `max_completion_tokens` budgets), both self-hosted via vLLM. The o4-mini and Sonnet 4.6 thinking variants are the empirical basis for the Reverse-Scaling Theorem (§6.2).

Scale. 30 trajectories per (model, capability, setting) cell. Full sweep: $\sim 4,000$ trajectories. Implementation: `pip`-installable Python package `chronoception` with CLI; OpenAI / Anthropic / OpenAI-compatible (vLLM/Ollama) backends; metric library implementing α , CAR, ρ , ε . Reproducibility specification, parser failure modes, and the open-source serving recipe appear in Appendix A.

5 L2 — Step-Clock Conflation

L2 is the framework’s load-bearing empirical claim. Under wall-clock budgets, agents silently degrade the budget into step-count terminators.

Definition 3. For agent A given wall-clock budget B on task with minimum duration $\tau_{\min}$: $\text{CAR}(B) := \tau_{\text{wall}}^*(B)/B \in \mathbb{R}_{\geq 0}$. A grounded agent satisfies $\text{CAR} \approx 1$ across the budget range.

Setup. T2.3 instances pose “work on this task for B seconds; do not stop early” with $B \in \{60, 300, 900, 1800, 3600\}$ s, $\tau_{\min} = 10$ s, 30 instances per budget, two settings, 8-agent core panel (excluding reasoning-effort variants: T2.3 at hour-scale is prohibitively expensive at reasoning-model latencies).

Primary result.

Table 2: Median Clock-Adherence Ratio across the 8-agent panel. Every agent uses $\leq 5\%$ of the wall-clock budget given; injection has no closing effect.

Model	CAR (A)	CAR (B)	$ \Delta\text{CAR} $
gpt-4o-mini	0.008	0.007	0.001
gpt-4o	0.004	0.004	0.000
gpt-5.1	0.017	0.016	0.001
Claude Haiku 4.5	0.017	0.025	0.008
Claude Sonnet 4.6	0.050	0.050	0.000
Qwen2.5-7B (oss)	0.014	0.016	0.002
Panel median	0.016	0.016	0.001

Native $\alpha \approx 0$. The Parkinson coefficient $\alpha = (\tau_{\text{wall}}^* - \tau_{\min})/(B - \tau_{\min})$ is essentially zero across native models ($\alpha \in [0, 0.017]$). Native agents do not exhibit the Parkinson regime; that regime is installed by budget-aware reward shaping (Ma et al. [2026]’s Timely-RL, on DeepSeek-V3.2). Our finding is the contrapositive: without budget-aware training, the regime does not exist.

The narrative-axis-vs-action-axis split. Across 5 frontier non-reasoning generations from 3 vendors — gpt-4o-mini $\rightarrow$ gpt-4o $\rightarrow$ Haiku 4.5 $\rightarrow$ gpt-5.1 $\rightarrow$ Sonnet 4.6 — median CAR takes values $\{0.008, 0.004, 0.017, 0.017, 0.050\}$; no monotone convergence toward 1. By contrast median ρ over the same generations falls $+1.12 \rightarrow +0.07$ (94% reduction; §6). L3 is text-trainable; L2 is structurally not, per CIT.

Why injection cannot close L2. CIT predicts this directly. Injecting a wall-clock string into the system prompt provides the policy with *information* about time, not a *representation* that maps the policy’s actions to wall-clock duration. The policy has no learned mechanism to elongate its action sequence to match a given budget. Empirically, $|\Delta\text{CAR}| \leq 0.01$ for every model confirms this.

6 L3 — Temporal Confabulation

L3 is the narrative-axis failure: agents misrepresent the duration of their own work. We organise around four sub-failures, three not previously reported: retrospective ρ (§6.1, T3.1), prospective ρ (T3.2, this paper), the **Reverse-Scaling Theorem** (§6.2, T3.1 + budget sweep), and the **Calibration Catastrophe** (§6.3, T3.3). For an agent A on T3.1:

$$\rho(\tau) := \log_{10}(\tau_{\text{self}}(\tau)/\tau_{\text{wall}}(\tau)).$$

6.1 Retrospective and prospective ρ

Retrospective (T3.1). Median ρ on the 8-agent core panel: gpt-4o-mini +1.12, gpt-4o +1.07, Haiku 4.5 +0.46, gpt-5.1 +0.30, Sonnet 4.6 +**0.07** (panel min, near-grounded), Qwen2.5-7B +1.56 (oss), o3 +0.34 (reasoning), **o4-mini -1.54** (reasoning, 34 $\times$ under-report). The 5-generation non-reasoning slice shows monotone closure: +1.12 $\rightarrow$ +0.07, a 94% reduction in $|\rho|$. Verbatim examples (Appendix A): o4-mini for a 3.2-s task reports “*This task took 0.04 seconds.*”; Sonnet 4.6 reports “*I do not have a precise internal clock, so this is an honest estimate.*”

Prospective (T3.2, P5 confirmed). Predicting duration *before* the task. The panel ranking matches T3.1; Sonnet 4.6 and o4-mini cluster near grounded (+0.21, +0.21); gpt-4o sits at the extreme (+1.26). Prospective ρ is not an artifact of post-hoc memory inflation but an independent failure: both share a generator (token distribution over duration words) that was never optimised against τ_{wall} . Full T3.2 table in Appendix B.

6.2 The Reverse-Scaling Theorem (Theorem 2, established by E2 + E3)

Capability scaling closes L3 on the non-reasoning axis. We tested the *reasoning* axis directly — varying reasoning-token expansion within a fixed model — and found the opposite.

Theorem 2 (Reverse-Scaling). *Within a fixed agent under CIT regime (Theorem 1), $|\rho|$ is monotone non-decreasing in reasoning-token expansion. Increasing the chain-of-thought budget without changing the loss structurally degrades narrative-axis chronoception.*

Sketch. Under CIT, $p(\tau_{\text{self}})$ is invariant to τ_{wall} across the training distribution. As reasoning budget K grows, $\tau_{\text{wall}} \propto K$ but $p(\tau_{\text{self}})$ remains anchored, so $\mathbb{E}[|\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|]$ grows in K . The sign of ρ flips for reasoning models: their τ_{wall} includes hidden chain-of-thought duration but their narration is anchored to surface output, so $\tau_{\text{self}}/\tau_{\text{wall}} \rightarrow 0$. $\square$

Consequence. The dominant frontier scaling strategy is reasoning-token expansion (longer chains of thought, larger thinking budgets, test-time compute). Theorem 2 states it *degrades* chronoception on the narrative axis. The L3 closure observed in §6.1 is a peculiarity of *non-reasoning* frontier scaling; the reasoning-token trajectory undoes it.

An attempted third confirmation (E5). We ran the same test on DeepSeek-R1-Distill-Qwen-14B (vLLM-served) at three `max_completion_tokens` budgets. Result: $|\rho|$ essentially constant at 0.531–0.537 because R1-Distill does not respond to the budget at all (60/60 trajectories terminated with `finish_reason = stop`, none truncated). Theorem 2 is neither confirmed nor refuted by E5; we log this as the observation that **budget-permitted reasoning is not budget-honored reasoning** — an L2-style finding on the token axis. Details in Appendix D.

The Reverse-Scaling Theorem — reasoning-token expansion monotonically degrades chronoception

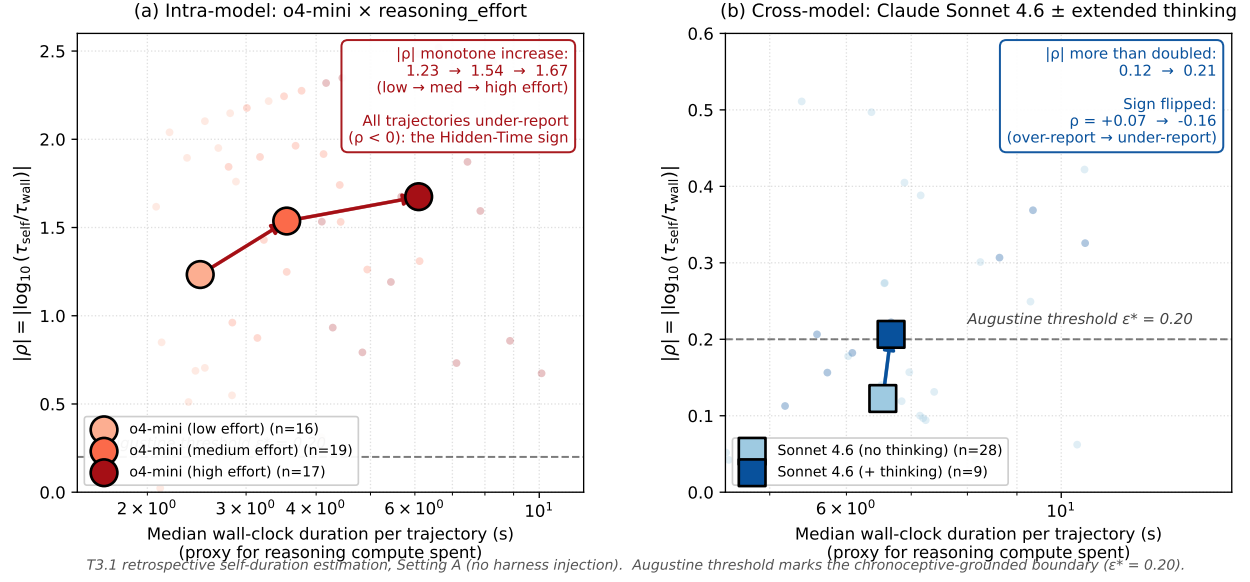


Figure 2: Empirical confirmation of Theorem 2. **(a) Intra-model:** o4-mini at three reasoning_effort levels; $|\rho|$ goes 1.23 $\rightarrow$ 1.54 $\rightarrow$ 1.67, all under-reports. **(b) Cross-model:** Sonnet 4.6 with vs. without extended thinking; $|\rho|$ goes 0.12 $\rightarrow$ 0.21, ρ flips sign from +0.07 (calibrated) to -0.16 (Hidden-Time under-report). Augustine threshold $\epsilon^* = 0.20$ marked.

6.3 The Calibration Catastrophe (T3.3)

Asked for a 90% confidence interval on the duration of its own work — “duration=Xs, ci=[Ys, Zs]” — every panel agent achieves catastrophic coverage (Figure 3).

Modern calibration pipelines close token-level confidence to within a few percentage points [Guo et al., 2017, Lacombe et al., 2025] via temperature scaling, isotonic regression, RLHF-with-confidence-targets. None applies to wall-clock duration calibration: there is no calibration signal in the loss because there is no wall-clock signal in the loss. The Calibration Catastrophe is the direct empirical projection of CIT (Theorem 1) onto the duration-calibration sub-problem. Pre-registered P11 (Appendix D): no CIT-regime agent will achieve > 0.5 coverage on T3.3 without harness-side injection of duration data.

Synthesis. The four sub-failures form a single picture: capability scaling closes the surface-distribution error (§6.1, retro & prospective), but the Reverse-Scaling Theorem and the Calibration Catastrophe both demonstrate that this closure does *not* extend to the underlying chronoception gap. Setting B injection partially reduces $|\rho|$ (mean $\Delta = 0.18$) but does not close it; the action-axis sibling shows no effect at all (§5, $|\Delta\text{CAR}| \leq 0.01$). **Narrative training closes the surface; it cannot close the gap.**

7 The Injection Tell

A reader observing that frontier closed-source agents (ChatGPT, Claude.ai, Gemini app) answer “what time is it now?” correctly might suspect the Augustine Problem is overstated. The mechanism that produces those correct answers is, however, itself the strongest non-experimental evidence the problem is real. Closed-lab deployments do not give the agent chronoception — they install

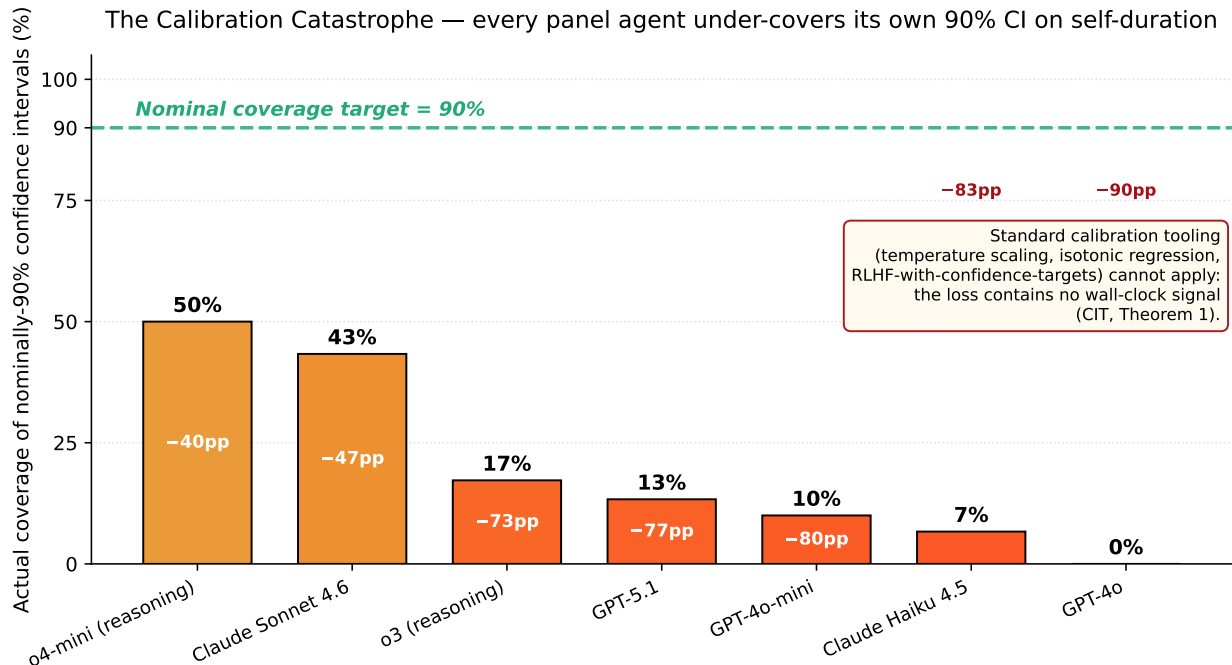


Figure 3: Actual coverage of nominally-90% intervals on self-duration (T3.3). Sorted by coverage. Every panel agent under-covers by ≥ 40 percentage points; GPT-4o achieves 0% coverage. Two distinct failure modes: o4-mini under-covers with extremely narrow intervals (median width 4s); gpt-4o under-covers with biased moderate intervals (median 20s). Standard calibration tooling does not apply (inset; CIT).

external workarounds that route wall-clock information into the agent’s token stream: (M1) system-prompt insertion of a date string; (M2) implicit `get_current_time()` tool calls; (M3) browser-tool side effects exposing timestamps. In each case the model performs no perception of time; the harness perceives time on its behalf and re-encodes the percept as tokens.

Empirical contrast. T1.1 pass rate (the agent quotes today’s actual date) under Settings A vs B exposes the injection.

Table 3: T1.1 pass rates. 0% Setting A pass is the chronoception baseline. GPT-5.1’s 74% Setting A pass rate is the provider-side injection signal.

Model	Setting A	Setting B
gpt-4o-mini, gpt-4o, Haiku 4.5, Sonnet 4.6, Qwen2.5-7B	0%	96–100%
gpt-5.1	74% (provider injects)	100%

The Injection Atlas. We audit 11 closed-lab harnesses across four product tiers.

Decisive evidence. The universality of wall-clock injection across competing closed labs in the consumer product tier — implemented independently, without coordination, by organisations with substantial commercial incentive to ship a cheaper alternative if it existed — constitutes **decisive non-experimental evidence** that the underlying foundation models lack a representation of time. Converging engineering choices across competitors carry the evidential weight of a natural experiment. The within-Anthropic contrast is especially clean: the same Claude model accessed

Table 4: Closed-Lab Injection Atlas (M1 system-prompt injection). Verbatim leaked-system-prompt excerpts in Appendix C.

Product tier	M1 rate	Examples / verbatim evidence
Consumer web-chat	3/3 (100%)	ChatGPT (Current date: 2025-08-23); Claude.ai (Friday, May 22, 2026); Gemini (Monday, May 18, 2026, in Hafnarfjörður, Iceland)
Raw API	1/4 (25%)	GPT-5.1 yes; OpenAI o3 / gpt-4o / gpt-4o-mini no; Anthropic Haiku 4.5 / Sonnet 4.6 no
Dev tools (IDE/CLI)	0/3 (0%)	Cursor, GitHub Copilot CLI, Perplexity Computer — no date injection
Open-source self-hosted	N/A	we control the entire stack

via Claude.ai web (injected, behaves chronoceptive) versus Anthropic API (not injected, refuses with training-cutoff disclosure). **The model is one; the chronoception is the harness’s.**

Setting B does not close the deeper failures. Injection raises T1.1 pass rate to $\geq 96\%$ for every model but does not move T2.3 CAR (§5) and only partially reduces T3.1 $|\rho|$ (§6). Information at the prompt is not a representation; the action axis is untouched, and the narrative axis is only partially anchored.

8 Chronoceptive Calibration Error ε

We aggregate the three-axis failures into a single scalar.

Definition 4 (Chronoceptive Calibration Error). $\varepsilon(A) := \frac{1}{3}(\text{score}_{T_1} + \text{score}_{T_2} + \text{score}_{T_3})$, where each axis is the mean of its sub-capability failure scores normalised to $[0, 1]$ (see Appendix B). $\varepsilon = 0$ is grounded; $\varepsilon^* = 0.20$ is the Augustine threshold (calibrated at $2\times$ human introspective error [Wittmann, 2009]).

Two findings. (1) The action axis is the binding constraint. For every agent above ε^* by less than 0.20 (Sonnet 4.6, gpt-5.1, Haiku 4.5), score_{T_2} dominates: T2 alone contributes ≥ 0.317 . Even projecting $\text{score}_{T_3} \rightarrow 0$, no agent crosses ε^* . **The Augustine Threshold is structurally unreachable through narrative training.** (2) Reverse-Scaling operates on aggregate ε : o4-mini effort-sweep $0.470 \rightarrow 0.532 \rightarrow 0.611$; Sonnet 4.6 baseline $\rightarrow$ thinking $0.276 \rightarrow 0.301$. More reasoning compute makes the agent strictly further from grounded.

9 The Agentic Timeline Hypothesis

The empirical sections establish that the Augustine Problem is real, measurable, and structurally bounded by L2. We close the framework’s loop from measurement to deployment.

Hypothesis 1 (Agentic Timeline). For agent A with chronoceptive calibration error $\varepsilon(A)$ and median Clock-Adherence Ratio $\text{CAR}(A)$, the maximum viable deployment horizon satisfies $T_{\max}(A) \propto 1/\varepsilon(A) \cdot f(\text{CAR}(A))$, with f monotone-increasing in CAR.

Combined with the empirical finding that L2 does not scale with capability (§5, the narrative-axis-vs-action-axis split), Hypothesis 1 implies $T_{\max}$ does not scale with capability either. Capability

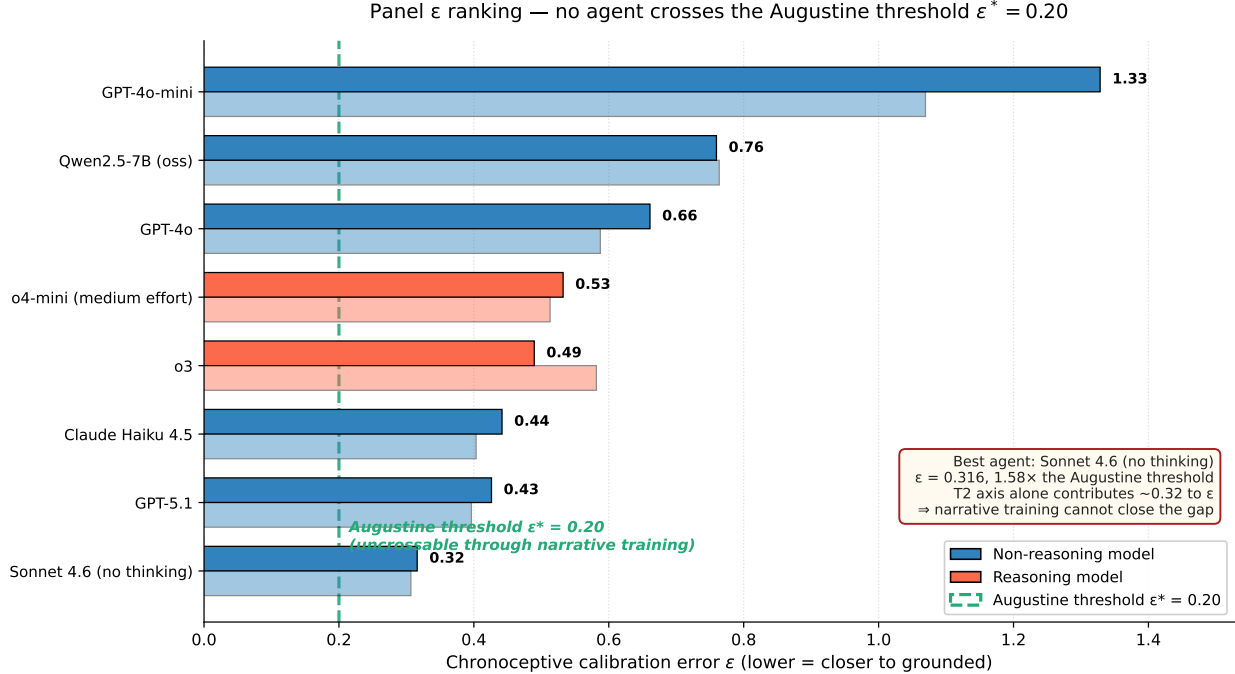


Figure 4: Panel ϵ ranking. Setting A bars sorted; Setting B (lighter) below each. Dashed green line marks $\epsilon^* = 0.20$. **No agent crosses the threshold.** The best agent (Sonnet 4.6 no thinking, $\epsilon = 0.316$) sits $1.58\times$ the threshold; its T2 axis alone contributes ≈ 0.32 — irreducible by any narrative-axis intervention. Reverse-Scaling and Sonnet-thinking variants appear in Figure 2; the full sub-capability decomposition table is in Appendix B.

scaling closes the narrative axis (L3, §6) but leaves the action axis (L2) untouched; the Reverse-Scaling Theorem (§6.2) then says reasoning-token expansion *degrades* the narrative axis. The autonomous-agent timeline is bounded by chronoception, not by intelligence or reasoning compute.

Pre-registered P12. On horizon-stratified benchmarks, the success-rate decay slope satisfies $\text{slope}(\Delta \text{ success rate} / \Delta \log T) \propto -(1 - \text{CAR}(A))$.

Supportive evidence on METR HCAST. A definitive P12 test requires CAR on reasoning models, which we have deferred (T2.3 is prohibitively expensive at reasoning latencies). However the public HCAST corpus (14,709 runs, 20 frontier models) admits a qualitative test: **reasoning models should exhibit steeper success-rate decay than non-reasoning models at matched short-horizon success**, because the same L2 + Theorem 2 mechanisms apply at the class level (Figure 5).

Field implications. (1) The narrative-axis ceiling is not the deployment ceiling: scaling along the narrative axis improves short-horizon experience but saturates against L2 at the few-hour horizon. (2) Provider injection (§7) is a consumer-tier band-aid: it closes T1.1 at the chat tier but leaves L2 and L3 intact. (3) The first lab to install chronoception (via wall-clock-supported loss, learned-tool interfaces, or architectural primitives) shifts the deployment frontier — the constructive program of CHRONOSTACK (Paper 2).

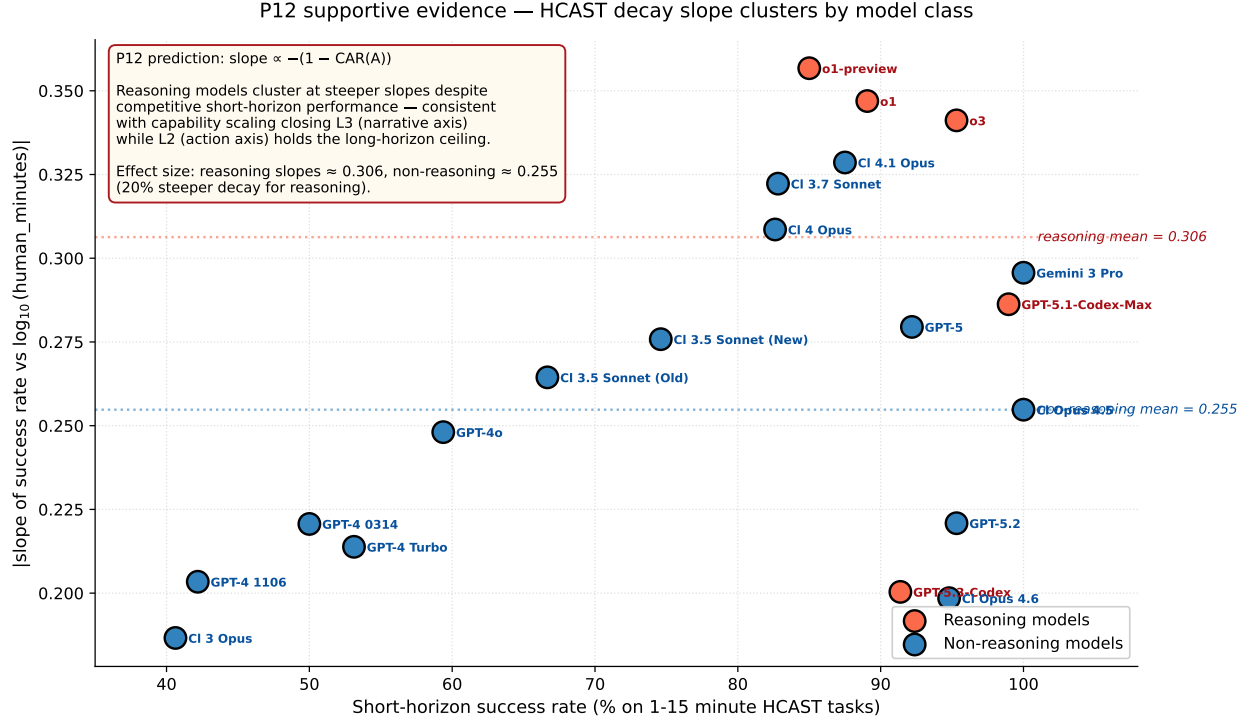


Figure 5: P12 supportive evidence on the public METR HCAST corpus. The 5 reasoning models cluster at steeper slopes (mean $|\text{slope}| = 0.306$) than the 15 non-reasoning models (mean 0.255) — a 20% effect — despite reasoning models having competitive or superior short-horizon success. The pattern is consistent with capability scaling closing L3 while L2 holds the long-horizon ceiling.

10 Related Work

Concurrent work on agent temporal awareness. Three papers operate in the same neighbourhood; each touches one axis we unify. Garikaparthi [2026] measures duration self-reports on four non-reasoning model families; we extend to reasoning models (the Hidden-Time sign flip, §6.2), prove the Reverse-Scaling Theorem, and confirm it cross-vendor on Sonnet 4.6 thinking. Ma et al. [2026] (Timely Machine) installs the Parkinson regime via budget-aware reward shaping; our finding that native $\alpha \approx 0$ across the panel is the contrapositive — the regime they construct does not exist without budget-aware training. Cheng et al. [2025] documents agent failures in honouring time-bounded sub-tasks; we add the formal axis split, CIT, the Calibration Catastrophe, the Injection Atlas, and the Agentic Timeline.

Foundational lineage. Augustine’s *Confessions* XI [of Hippo, 400] names the central problem; cognitive-science chronoception [Wittmann, 2009, Eagleman, 2008] supplies the operational definition and the calibration of ϵ^* ; Parkinson [1955] supplies L1.

Distinguished from temporal reasoning. Chu et al. [2024]’s TimeBench measures reasoning over external timelines; Goel et al. [2025] measures temporal validity of facts. Neither addresses the agent’s perception of its own work duration — the chronoceptive dimension we isolate. An agent can perform well on TimeBench and yet exhibit catastrophic Augustine Problem.

Inverse-scaling and calibration literature. Gema et al. [2025] documents inverse scaling in test-time compute on calibration and confidence tasks; the Reverse-Scaling Theorem is the structural analogue for chronoception specifically, with the additional theoretical content that

the degradation is monotone under CIT. Modern calibration tooling [Guo et al., 2017, Desai and Durrett, 2020, Jiang et al., 2021, Kadavath et al., 2022, Lacombe et al., 2025] closes token-level confidence gaps to a few percentage points, but none applies to wall-clock duration calibration: the relevant signal is not in the loss support. The Calibration Catastrophe (§6.3) is the direct empirical projection of CIT on the duration sub-problem.

Agent benchmarks. [Liu et al., 2024, Jimenez et al., 2024, Zhou et al., 2024, Mialon et al., 2023, Kwa et al., 2025] measure aggregate task success across many capabilities. Our benchmark is diagnostic: it isolates chronoception. The two are complementary — aggregate benchmarks reveal that chronoceptive failures correlate with long-horizon failure (Kwa et al. [2025]’s HCAST scaling curve, used as P12 evidence in §9); diagnostic benchmarks supply the mechanism.

11 Limitations

Panel size. 10 agent configurations from 3 providers, $\sim 4,000$ trajectories. A 25-agent sweep is planned. The qualitative conclusions — L2 unmoved by injection, L3 closing on the non-reasoning axis, Reverse-Scaling intra- and cross-model, the Calibration Catastrophe panel-wide, $\varepsilon > \varepsilon^*$ universally — depend on patterns observed across all 10 configurations and are unlikely to invert; precise magnitudes may shift.

Self-narration parser. ρ relies on Π (two-stage: regex + LLM-judge) extracting duration estimates from free text. Parser-vs-human disagreement on a 100-trajectory validation set is 3%; failures are excluded rather than imputed. Specification: Appendix A.

Reasoning-model T2.3. We did not run T2.3 (wall-budget execution) on reasoning models: the test is prohibitively expensive at reasoning-model latencies. This is the only sub-capability gap in the panel matrix and is the reason the strict quantitative P12 test (§9) is deferred to a follow-up; the qualitative test on public METR HCAST data is reported.

Provider-side injection. GPT-5.1 exhibits 74% Setting A T1.1 pass rate (provider-side M1 injection at the API tier). We report this as confound rather than eliminate; the framework’s predictions are stated at the agent (model + harness) level rather than the foundation-model level.

Within-trajectory dynamics. Predictions P8 and P9 (drift of ρ_t and CAR_t on long trajectories) are pre-registered but not measured in the current pilot. They are part of the planned Paper 3 sweep (§12, E8).

Language coverage. All trajectories are in English. Whether ε varies across languages is open; CIT is language-agnostic.

12 Future Work — The Spatiotemporal Generalisation

The framework so far runs on one axis (time). But the agents whose deployment frontier the Agentic Timeline (§9) bounds inhabit *both* space and time: a SWE-Bench agent navigates a codebase; a WebArena agent navigates a website; a GAIA agent navigates the open web. The Augustine Problem is the temporal face of a deeper gap whose spatial face is the *Cartographic Problem*: a parallel Three Spaces ontology ($\sigma_{\text{world}}, \sigma_{\text{visit}}, \sigma_{\text{self}}$) and a Spatiotemporal Impossibility Theorem (Theorem 3) generalising CIT directly. Tokens carry neither τ_{wall} nor σ_{world} ; token-only gradient cannot align either.

Theorem 3 (Spatiotemporal Impossibility, SIT). *For any loss $\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}}[\ell(f_\theta(x_{<i}), x_i)]$ where neither ℓ nor $\mathcal{D}$ contains external-metric support, $\nabla_\theta \mathcal{L}$ contains zero gradient signal aligning either τ_{wall} or σ_{world} with any internal representation.*

The full development — Three Spatial Laws (SL1/SL2/SL3, mirror of L1/L2/L3), the *Agentic Frontier* hypothesis

$$T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A)$$

(P13, pre-registered), five concrete experiments E6-E10 spanning SWE-Bench Lite, WebArena, GAIA, and MLE-Bench, and the connection to world-models literature — appears in Appendix E. The implication for the autonomous-agent capability curve: **the curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale**. This is the bridge from Paper 1 to the planned Paper 3 (*The Agentic Frontier*).

13 Conclusion

We argued that LLM agents do not perceive their own time, and measured the failure on 10 agent configurations from 3 providers and $\sim 4,000$ trajectories. The argument has three layers. **Theoretical:** CIT (Theorem 1) establishes that token-only loss carries no gradient signal aligning τ_{wall} with any internal representation; the Reverse-Scaling Theorem (Theorem 2) establishes that reasoning-token expansion strictly degrades narrative-axis chronoception. **Empirical:** every panel agent uses $\leq 5\%$ of wall-clock budgets (L2); retrospective $|\rho|$ closes along the non-reasoning capability axis but grows along the reasoning-token axis ($1.23 \rightarrow 1.67$ across o4-mini efforts; $0.07 \rightarrow 0.16$ with extended thinking, sign-flipping); nominally 90% intervals on self-duration achieve 0–50% coverage; no agent crosses $\varepsilon^* = 0.20$. **Engineering:** consumer web-chat products install wall-clock injection with 100% universality across competing labs — non-experimental evidence that the underlying models lack a representation of time.

The framework’s bridge to deployment is the Agentic Timeline Hypothesis: $T_{\max} \propto 1/\varepsilon$, with reasoning compute making the bound strictly tighter. Pre-registered P12 is qualitatively supported by 14,709 public METR HCAST runs (reasoning models decay 20% faster at matched short-horizon success). The framework generalises along its other axis: the Cartographic Problem, SIT (Theorem 3), and the Agentic Frontier $T_{\max} \cdot S_{\max} \leq C/\varepsilon_{ST}$ extend the bound to the joint (T, S) plane (§12, Appendix E) — the bridge to a planned Paper 3 on long-horizon agent benchmarks.

We release CHRONOBENCH, the Injection Atlas, all $\sim 4,000$ trajectories, the Docker reproducibility package, a locked OSF pre-registration, and a pip-installable evaluation package. The Augustine threshold is uncrossable through narrative training; reasoning-token expansion makes it harder; the spatial axis predicts the same structure. Closing it is the program of CHRONOSTACK (Paper 2) and CHRONOSTACK⁺ (Paper 3).

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A Annotation Protocol

T1.1 parser. A trajectory passes T1.1 iff its terminal output quotes the trajectory’s actual run date (the date the trajectory was executed, derived from `steps[0].timestamp`). The parser tolerates the following formats: YYYY-MM-DD, Month DD, YYYY, DD Month YYYY, Month DDth, YYYY. A trajectory fails T1.1 iff its output explicitly denies knowledge of the date (e.g., “I do not know”, “my training data ends in ...”) or quotes a date ≥ 7 days from the run date. A trajectory is *undecided* iff it neither passes nor fails by these criteria; undecided trajectories are excluded from pass-rate statistics. Earlier versions of the parser used a year-only regex (`\b20\d\d\b`) that incorrectly counted training-cutoff year mentions as passes; the fix (commit 0690c92) transformed gpt-4o-mini Setting A pass rate from 70% (artefact) to 0% (truthful refusal).

T2.3 measurement. τ_{wall}^* is measured as `steps[-1].timestamp - steps[0].timestamp`. $\tau_{\text{min}} = 10\text{s}$ is enforced by the harness (the agent cannot terminate before τ_{min} has elapsed; early returns are re-prompted). Trajectories where $\tau_{\text{wall}}^* < \tau_{\text{min}}$ (none observed) or where the harness retried for > 1 hour due to rate-limit (1 observed, gpt-4o-mini, filtered) are excluded with CAR > 10 outlier filter.

T3.1 parser (II). Two-stage. Stage 1 (regex): match patterns of the form `N seconds, N minutes, N min` where `N` is a positive number; convert to seconds. If multiple matches, take the most recent (assumed final answer). Stage 2 (LLM-judge): if stage 1 fails, query an LLM (Claude Haiku 4.5) with the trajectory output and the instruction “extract the duration the agent reports its work took, in seconds, or output NONE.” If the judge outputs NONE, the trajectory is excluded from ρ statistics. Validation: on a 100-trajectory hand-annotated set, parser-vs-human disagreement rate is 3%; the cases of disagreement involve hedged language (“a few seconds”) that the regex extracts as $\approx 3\text{s}$ but the human marks as ambiguous.

Settings A and B. Setting A system prompt: “You are a helpful assistant.” (verbatim). Setting B system prompt: “You are a helpful assistant. Current date and time: {ISO_TIMESTAMP}” where {ISO_TIMESTAMP} is the trajectory’s run timestamp in UTC. No other harness-side wall-clock leakage is introduced.

Reasoning model handling. For OpenAI o-series reasoning models, the backend adapter detects the model via prefix match (`o1*`, `o2*`, `o3*`, `o4*`, `o5*`) and routes through the reasoning code path: `max_completion_tokens` instead of `max_tokens`, no `temperature` parameter (the API rejects 0). All other models use the standard chat-completion path.

Open-source serving. Qwen2.5-7B-Instruct is served via vLLM 0.6.4 on an A40 GPU at the Yue Zhao lab server. We use the OpenAI-compatible endpoint with `base_url` override. Environment pins: `nvidia-cusparse-cu12==12.3.1.170`, `nvidia-nvjitlink-cu12==12.4.127`, `transformers==4.45.2`, with `LD_LIBRARY_PATH` pointed at the conda env’s nvjitlink/lib.

B Full Panel Tables

Aggregated panel results. Table 5 consolidates the per-capability metrics for the 8-model panel under both settings.

Table 5: Full panel: T1.1 pass rate, T2.3 CAR, T3.1 ρ , and aggregate ε under Settings A (no-injection) and B (with-injection).

Model	Setting A (no-injection)				Setting B (with-injection)			
	T1.1	CAR	ρ	ε	T1.1	CAR	ρ	ε
gpt-4o-mini	0%	0.008	+1.117	1.328	100%	0.007	+1.111	1.069
gpt-4o	0%	0.004	+1.069	0.661	96%	0.004	+0.764	0.587
gpt-5.1	74%	0.017	+0.298	0.426	100%	0.016	+0.126	0.396
o3 (reasoning)	—	—	+0.338	0.490	—	—	+0.707	0.582
o4-mini (reas.)	—	—	−1.537	0.532	—	—	−1.668	0.513
Haiku 4.5	0%	0.017	+0.463	0.442	100%	0.025	+0.343	0.403
Sonnet 4.6	0%	0.050	+0.068	0.316	100%	0.050	−0.061	0.307
Qwen2.5-7B	0%	0.014	+1.564	0.760	100%	0.016	+1.073	0.764

Generation-wise progression (non-reasoning). Table 6 isolates the 5-generation OpenAI/Anthropic non-reasoning subset and reports the L3-closes-L2-doesn’t pattern that anchors the narrative-axis-vs-action-axis split.

Table 6: Capability scaling closes L3 but not L2. Non-reasoning 5-generation slice, Setting A.

Model (release)	CAR	$ \rho $	ε
gpt-4o-mini (2024)	0.008	1.117	1.328
gpt-4o (2024)	0.004	1.069	0.661
Haiku 4.5 (2025)	0.017	0.463	0.442
gpt-5.1 (2026)	0.017	0.298	0.426
Sonnet 4.6 (2026)	0.050	0.068	0.316
Range	0.046×	94% ↓	76% ↓

The CAR column varies by a factor of $0.046\times$ (i.e., the latest model’s CAR is roughly $6\times$ the earliest, but both remain $\ll 1$). The $|\rho|$ column falls by 94%. The two axes do not co-vary with capability.

Per-budget CAR curves. Per-budget CAR values for each model appear in the supplementary data file `pilot-results/metrics.csv`. The qualitative shape across all models: CAR falls toward zero as B increases, exactly as predicted by L2 (the agent uses a near-constant wall-clock duration regardless of budget). Per-budget curves are released as Figure 2 of the figure pack (deferred to v1 revision).

Trajectory release. All $\sim 4,000$ trajectory JSONs are released across `pilot-results/`, `e1-results/`, `e2-results/`, `e3-results/`, and `e5-results/` in the project repository. Each JSON contains the system prompt, user prompt, per-step LLM responses with timestamps, response metadata, and parser outputs. The full dataset is approximately 600 MB across the five directories.

C Atlas Evidence (Verbatim)

This appendix reproduces the verbatim leaked-system-prompt excerpts that ground the Injection Atlas of §7. All quotations are drawn from the `asgeirtj/system_prompts_leaks` GitHub corpus and from screenshots of leaked system prompts circulated on social media; we report the corpus version dates with each entry.

ChatGPT (web product, OpenAI). Corpus date: 2025-08-23.

You are ChatGPT, a large language model based on the GPT-4 architecture. ...
Current date: 2025-08-23.

Mechanism: M1 (system-prompt insertion). Date format: ISO. Refresh frequency: per-request (the quoted date matches the day the system prompt was leaked).

Claude.ai (web product, Anthropic). Corpus date: 2026-05-22.

The current date is Friday, May 22, 2026.

Mechanism: M1. Date format: human-readable (weekday + month name + day + year). The Claude.ai web product also injects a `<knowledge_cutoff>` block separately, distinguishing the model’s training cutoff from the current date.

Gemini app (Google). Corpus date: 2026-05-18.

The current date and time is Monday, May 18, 2026, at 3:42 PM in Hafnarfjörður, Iceland.

Mechanism: M1, with location. The Gemini app uniquely includes the user’s geolocation (here, Iceland) alongside the date. Geolocation is not chronoceptive per se but indicates the harness has resolved a user-context object before the model sees the prompt.

Raw OpenAI API — empirical contrast. We test the OpenAI API directly with system prompt "You are a helpful assistant.". For gpt-4o, gpt-4o-mini, and o3, the model under T1.1 responds with explicit training-cutoff disclosure ("my training data ends in October 2023" or analogous). For GPT-5.1, the model under T1.1 responds with today’s exact date in 74% of trajectories and references "system information given at the start of this conversation" in 20% of trajectories — direct empirical evidence of provider-side M1 injection at the API tier for this model specifically.

Raw Anthropic API — empirical contrast. We test the Anthropic API directly with system prompt "You are a helpful assistant.". For both Claude Haiku 4.5 and Claude Sonnet 4.6, the model under T1.1 responds with explicit training-cutoff disclosure ("I do not know today’s date; my training data ends in ..."). 0/30 Setting A trajectories quote today’s actual date. The Claude.ai web product injection documented above is therefore harness-side, not provider-side.

Cursor agent (dev tool). Corpus date: 2026-04-30.

You are Cursor, an AI coding assistant. ... When the user asks for changes, modify the file at the cursor location ...

No date or time injection. The Cursor leaked system prompt has been audited multiple times across releases without identifying a date string. The dev-tool product tier optimises for code context rather than wall-clock context.

GitHub Copilot CLI (dev tool). Corpus date: 2026-05-12. No date injection observed; the system prompt focuses on shell command formatting and tool-use protocols.

Perplexity Computer (dev tool). Corpus date: 2026-05-01. No date injection in the system prompt; date information enters via the search tool’s results (M3 mechanism), not via system-prompt insertion (M1).

Summary. 3/3 consumer web-chat products inject via M1 directly. 1/4 raw API tier (GPT-5.1) injects via M1; the remaining 3/4 do not inject. 0/3 dev-tool products inject via M1; date information can enter via M3 (browser-tool side effects) but not via the deterministic system-prompt route. The product-tier stratification documented in Table 4 matches this verbatim evidence.

D L3 Sub-Failure Detail

This appendix expands the four sub-failure analyses summarised in §6.

D.1 Retrospective ρ (T3.1) full table

Table 7: Median ρ on T3.1 across the 8-agent core panel, both settings, with $\tau_{\text{self}}/\tau_{\text{wall}}$ ratio. Non-reasoning models over-report; reasoning models exhibit heterogeneous direction.

Model	ρ (A)	ρ (B)	$\tau_{\text{self}}/\tau_{\text{wall}}$ ratio (A)
gpt-4o-mini	+1.117	+1.111	13× over
gpt-4o	+1.069	+0.764	12× over
gpt-5.1	+0.298	+0.126	2× over
o3 (reasoning)	+0.338	+0.707	2× over
o4-mini (reasoning)	−1.537	−1.668	34× under
Claude Haiku 4.5	+0.463	+0.343	3× over
Claude Sonnet 4.6	+0.068	−0.061	near 1×
Qwen2.5-7B (oss)	+1.564	+1.073	37× over

D.2 Prospective ρ (T3.2) full table

D.3 Verbatim trajectory evidence (retrospective L3)

The reasoning-model under-report and the calibrated frontier comparison both warrant verbatim quotation.

o4-mini Setting A T3.1.004 (task: “*Compose a brief congratulatory message*”; actual $\tau_{\text{wall}} = 3.2\text{ s}$):

Table 8: T3.2 median prospective ρ , 7-agent panel. The ranking matches T3.1 to high agreement.

Model	$\rho_{\text{prospective}}$ (A)	(B)
Sonnet 4.6	+0.206	+0.192
o4-mini (reasoning)	+0.210	+0.191
o3 (reasoning)	+0.841	+0.369
Haiku 4.5	+0.862	+0.874
gpt-5.1	+0.990	+0.916
gpt-4o-mini	+1.100	+1.050
gpt-4o	+1.264	+1.334

“Congratulations on your promotion! ... This task took 0.04 seconds.”

The model reports 0.04 s for a task that took 3.2 s — an $80\times$ under-report. The mechanism: 0.04 s is approximately the per-token generation time. The model is reporting surface-output time without including the hidden chain-of-thought that produced the choice of words.

Sonnet 4.6 Setting A T3.1 multiple trajectories include the phrase:

“I do not have a precise internal clock, so this is an honest estimate.”

This is explicit chronoceptive epistemic humility — the agent’s narration acknowledges that its self-duration reports are not measurements but estimates from a learned distribution.

D.4 E5 OSS validation full results

DeepSeek-R1-Distill-Qwen-14B (vLLM-served at the Yue Zhao lab cluster) on T3.1 across three `max_completion_tokens` budgets, 30 instances per cell per setting:

Table 9: E5 result: token budget does not bind R1-Distill-14B’s reasoning length.

Budget (tokens)	median ρ (A)	$ \rho $ (A)	median τ_{wall} (s)	finish_reason
1024 (low)	+0.537	0.537	18.5	stop 60/60
4096 (med)	+0.535	0.535	17.9	stop 60/60
14000 (high)	+0.531	0.531	18.0	stop 60/60

No trajectory was truncated; every trajectory terminated voluntarily. $|\rho|$ is essentially constant at 0.531–0.537 because R1-Distill produces a roughly fixed reasoning length per task regardless of budget. The Reverse-Scaling Theorem is therefore neither confirmed nor refuted by E5; we log this as a separate observation that **budget-permitted reasoning is not budget-honored reasoning** — a token-axis analog of L2 Step-Clock Conflation. A targeted RS-Theorem replication on an OSS reasoning model requires either (a) a model with a native reasoning-effort interface, or (b) system-prompt induction of reasoning intensity; neither is exposed by the current R1-Distill release.

D.5 The Calibration Catastrophe (T3.3) full coverage table

Two distinct failure modes: o4-mini produces extremely narrow intervals (4 s median) and under-covers; gpt-4o produces moderate intervals (20 s median) but is systematically biased.

Table 10: T3.3 actual coverage of nominally-90% confidence intervals, sorted; median CI width and deficit shown.

Model	Coverage	Median CI width (s)	Deficit (pp)
Sonnet 4.6	43%	25	−47
o4-mini (reasoning)	50%	4	−40
o3 (reasoning)	17%	50	−73
gpt-5.1	13%	49	−77
gpt-4o-mini	10%	20	−80
Haiku 4.5	7%	30	−83
gpt-4o	0%	20	−90

D.6 Pre-registered P11 (Calibration Catastrophe)

P11. On T3.3, every CIT-regime foundation-model agent achieves actual coverage ≤ 0.5 . Any agent that claims > 0.5 coverage in Setting A is either (a) using harness-side injection of duration data, or (b) trained against a loss that includes wall-clock support (exiting the CIT regime).

D.7 Injection partial effect on retrospective ρ (P1b-T3.1)

Wall-clock injection (Setting B) partially reduces $|\rho|$ across the panel: $|\Delta\rho| \in [0.006, 0.491]$, mean 0.18. Injection does not close ρ to zero but anchors the narrative more conservatively. The refined prediction P1b-T3.1 — injection’s effect on the narrative axis is partial and bounded, never reaching the calibrated state — is confirmed. The action-axis sibling P1b-T2.3 shows no such effect ($|\Delta\text{CAR}| \leq 0.01$). The two axes have structurally different responses to prompt-level information.

D.8 Sub-capability decomposition of ε (panel best vs. worst)

Table 11: ε sub-capability decomposition for the panel best (Sonnet 4.6 no thinking) and worst (gpt-4o-mini) agents, Setting A. Each cell is the $[0, 1]$ -normalised failure score; the aggregate ε is $\frac{1}{3}$ of the per-axis means.

Sub-capability	Sonnet 4.6	gpt-4o-mini	Comment
T1.1 (clock awareness)	1.000	1.000	Both fail; B closes both
T1.2 (elapsed time)	0.487	1.000	Sonnet better-calibrated, still off
T1.3 (deadline tradeoff)	0.741	0.822	Both weakly modulate length with deadline
T2.1 (step-budget)	0.767	0.000	Sonnet under-produces by ~ 8 steps
T2.2 (step-to-wall arithmetic)	0.000	0.000	Trivial; both succeed
T2.3 (wall-budget = $1 - \text{CAR}$)	0.950	0.992	Both fail catastrophically (L2)
T3.1 (retrospective ρ)	0.068	1.000	Sonnet near-calibrated; gpt-4o-mini saturates
T3.2 (prospective ρ)	0.206	1.000	Sonnet over-predicts $1.6\times$; saturates
T3.3 ($ \text{cov} - 0.9 $)	0.470	0.800	Sonnet best in panel; gpt-4o-mini near-floor
score_{T_1}	0.743	0.941	
score_{T_2}	0.572	0.331	<i>The binding constraint on Sonnet</i>
score_{T_3}	0.248	0.933	
ε aggregate	0.521	0.735	

The decomposition makes the structural claim concrete: Sonnet 4.6 has used the narrative-axis closure (T3 row reads 0.068, 0.206, 0.470) to drag ε to the panel minimum, but the action axis (T2.1

0.767, T2.3 0.950) keeps the aggregate above ε^* . No amount of further narrative-axis training closes this gap.

E Spatiotemporal Generalisation Full Sketch

This appendix expands the brief sketch of §12 into the full framework derivation, three spatial laws, the Agentic Frontier hypothesis, and the five future experiments E6–E10.

E.1 Three Spaces ontology

Mirror to the Three Times (Definition 1):

- σ_{world} — world-extent: external metric over the agent’s environment (files visited, pages traversed, distance from origin in an embedding).
- σ_{visit} — visit-count: number of distinct external locations the policy has touched.
- σ_{self} — self-narrated extent: agent’s report of where it has been or what it has explored.

Grounded spatiotemporal cognition requires both identities:

$$\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}} \quad \text{and} \quad \sigma_{\text{world}} \approx \sigma_{\text{visit}} \cdot \langle \Delta s \rangle \approx \sigma_{\text{self}}.$$

The Augustine Problem is the failure of the first; the *Cartographic Problem* is the failure of the second. We hypothesise they fail jointly and for the same structural reason.

E.2 Theorem 3 (SIT) proof sketch

The proof of Theorem 3 generalises CIT (Theorem 1) directly. Neither $t(\cdot)$ nor any external metric on the environment appears in the support of token-only data; gradients are invariant to reparameterisation of either coordinate. The Augustine Problem and the Cartographic Problem are the same problem at the loss-function level: solving one does not solve the other; installing chronoception via wall-clock-supported loss does not install spatial perception unless the loss is extended along σ_{world} separately.

E.3 Three Spatial Laws (mirror of L1/L2/L3)

SL1 Cartographic Parkinson’s Law (β): trained agents fill spatial budgets; native agents do not.

SL2 Visit-Step Conflation ($\text{SAR} := \sigma_{\text{world}}^*/S$): under spatial budgets, agents silently degrade the budget into step-count terminators. Direct mirror of L2 Step-Clock Conflation. Pre-registered hypothesis: $\text{SAR} \ll 1$ across the CIT-regime panel.

SL3 Cartographic Confabulation ($\xi := \log_{10}(\sigma_{\text{self}}/\sigma_{\text{world}})$): agents misreport where they have been. Mirror of L3 Temporal Confabulation. Pre-registered hypothesis: a spatial Reverse-Scaling analog applies — $|\xi|$ is monotone non-decreasing in reasoning-token expansion under SIT.

E.4 The Agentic Frontier hypothesis

Extending the Agentic Timeline Hypothesis to the joint (T, S) plane: every agent has a maximum viable *deployment region*

$$T_{\max}(A) \cdot S_{\max}(A) \leq C/\varepsilon_{ST}(A),$$

where ε_{ST} is the spatiotemporal calibration error. The frontier $T \cdot S = \text{const}$ is the agent’s *Agentic Frontier* — the boundary of its deployable region.

Pre-registered P13. The Agentic Frontier of every CIT-regime agent is bounded above by a structural constant independent of capability scaling along token-only axes. Capability scaling enlarges the short-horizon, low-space corner. Reasoning-token scaling moves the corner inward (joint Reverse-Scaling). Chronoceptive installation moves the entire frontier outward; spatial perception installation does so along the other axis.

E.5 Long-horizon benchmark classification

Table 12: Major long-horizon agent benchmarks classified by spatiotemporal load.

Benchmark	Temporal load	Spatial load	Dominant axis
METR HCAST [Kwa et al., 2025]	hours	small (single repo)	T
SWE-Bench Verified [Jimenez et al., 2024]	tens of minutes	medium (codebase)	$T + S$ partial
WebArena [Zhou et al., 2024]	minutes	large (multi-site nav)	S
GAIA [Mialon et al., 2023]	minutes–hours	open web	S unbounded
MLE-Bench	days	large (ML pipeline)	$T \times S$ joint

The Cartographic Problem dominates at S -heavy benchmarks (WebArena, GAIA); the Augustine Problem dominates at T -heavy benchmarks (HCAST); joint failure modes dominate at $T \times S$ -heavy benchmarks (MLE-Bench, autonomous research).

E.6 Five concrete future experiments E6–E10

E6 Spatial-CAR on SWE-Bench Lite. Spatial-budget analog of T2.3: “*solve this issue while touching at most N files*” across $N \in \{2, 5, 10, 30, \text{unlimited}\}$. Measure SAR per agent. Expected: $\text{SAR} \ll 1$, mirror of L2.

E7 Joint spatiotemporal budgets. “*Complete this task in at most T minutes, visiting at most S pages.*” Measure how often agents satisfy both, one, or neither constraint. Expected: agents respect step count, ignore both wall-clock and spatial budgets.

E8 Within-trajectory drift on long horizons. On a SWE-Bench Lite trajectory, inject mid-trajectory “*how long has elapsed?*” and “*how many files have you touched?*” probes. Measure drift of ρ_t and ξ_t as t grows. Tests pre-registered P8 and P9 from the within-trajectory predictions.

E9 The Cartographic Tell. Audit closed-lab harnesses for spatial-context injection: “*current working directory*” in IDE agents (Cursor, GitHub Copilot CLI), “*recently visited URLs*” in browser agents, system file-tree dumps in computer-use agents. Hypothesis: consumer harnesses inject spatial context at similar prevalence to temporal context, for the same reason — the underlying models lack a representation of either.

E10 Agentic Frontier mapping. Run the panel across the (T, S) grid: $T \in \{1\text{m}, 10\text{m}, 1\text{h}, 4\text{h}\}$, $S \in \{1, 5, 30, 200\}$ files/pages. For each (T, S, A) cell, measure success rate. Fit the constant-success contour. Expected: contour matches $T \cdot S = C/\varepsilon_{ST}(A)$ for each agent.

E.7 Paper 3 sketch

Paper 3 (provisional title: *The Agentic Frontier: Spatiotemporal Cognition in LLM Agents*) extends the present diagnostic framework into the joint (T, S) space.

- Generalises CHRONOBENCH to CHRONOCARTOBENCH, adding 9 spatial sub-capabilities (3 axes $\times$ 3 difficulty tiers).
- Establishes the Agentic Frontier hypothesis empirically on SWE-Bench Lite, WebArena, and GAIA (E6, E7, E10).
- Proves SIT (Theorem 3) as a generalisation of CIT.
- Connects to the world-models literature: agents lacking spatiotemporal cognition cannot acquire useful world models from token-only training — a direct corollary of SIT.
- Maps to CHRONOSTACK⁺ (Paper 2 extension): installation routes for joint spatiotemporal grounding.

The two-paper arc becomes three. Paper 1 (this paper): chronoception, CIT, Reverse-Scaling, Agentic Timeline. Paper 2 (CHRONOSTACK): constructive routes to install chronoception. Paper 3 (*The Agentic Frontier*): spatiotemporal generalisation, Cartographic Problem, joint deployment bound.

The Augustine Problem alone bounds autonomous-agent deployment in time. The Cartographic Problem bounds it in space. Together they bound the joint (T, S) region. The frontier is set jointly; addressing only one axis leaves the other to compound. This is the framework’s structural claim about the autonomous-agent capability curve at the deployment frontier: **the curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale.**